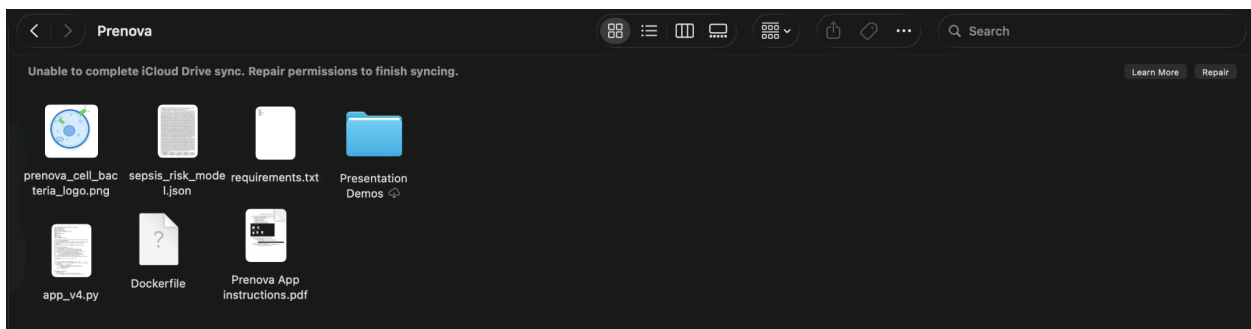


Prenova application instructions to Run

- 1) Install or make sure Docker Desktop application is installed on on your computer
 - a) <https://docs.docker.com/desktop/>
- 2) Download “Prenova” Folder that contains the following
 - a) Docker File = “Dockerfile”
 - i) Builds Container
 - b) Requirements = “requirements.txt”
 - i) Has list of libraries required for running Code
 - c) Application = “app_v4.py”
 - i) Application that has models and builds website
 - d) Logo = “prenova_cell_bacteria_logo.png”
 - i) Logo for website
 - e) Model Data = “sepsis_risk_model.json”
 - i) Data that model uses to run
 - f) Instructions = “Prenova App Instructions.pdf”
 - i) Instructions on how to run
 - g) Testing Files = “Presentation Demos” Folder
 - i) Actual Patient Data to be used to test application



- 3) Open up computer terminal (Mac) and load prenova folder
 - a) Read into Prenova Folder
 - i) Type in “cd” followed by location Prenova folder is saved
(1) For example:

```
GiancarloCedano@Giancarlos-MacBook-Pro ~ % cd /Users/GiancarloCedano/Desktop/Prenova
```

- 4) Build Docker Container for Prenova application
 - a) Type “ docker build --no-cache -t prenova-app . “ exactly has written here
 - i) This will build the application as shown below

```

GiancarloCedano@Giancarlos-MacBook-Pro Prenova % docker build --no-cache -t prenova-app .
[+] Building 136.8s (12/12) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                1.6s
=> => transferring dockerfile: 279B                                1.5s
=> [internal] load metadata for docker.io/library/python:3.11-slim 5.9s
=> [internal] load .dockerignore                                    0.0s
=> => transferring context: 2B                                       0.0s
=> [1/7] FROM docker.io/library/python:3.11-slim@sha256:6d85378d88a19cd4 0.0s
=> => resolve docker.io/library/python:3.11-slim@sha256:6d85378d88a19cd4 0.0s
=> [internal] load build context                                    0.0s
=> => transferring context: 84.11kB                                   0.0s
=> CACHED [2/7] WORKDIR /app                                       0.0s
=> [3/7] COPY requirements.txt .                                    0.0s
=> [4/7] RUN pip install --no-cache-dir -r requirements.txt        81.6s
=> [5/7] COPY app_v4.py .                                          0.1s
=> [6/7] COPY sepsis_risk_model.json .                             0.0s
=> [7/7] COPY prenova_cell_bacteria_logo.png .                    0.0s
=> exporting to image                                              47.5s
=> => exporting layers                                              39.1s
=> => exporting manifest sha256:76ee2e1d1bfb29bbfa541e2eb68231caa4436b46 0.0s
=> => exporting config sha256:6cc2e132d0cb03f9fbb1285b7e6cdc5f8025f2e2a2 0.0s

```

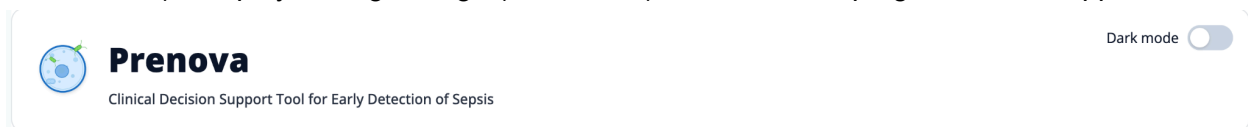
- 5) Once application has been built, run application on local host website
 - a) Type in “ docker run --rm -p 8005:8003 prenova-app”
 - b) This will load the application onto local website in the 8005 port
 - i) <http://localhost:8005/>
 - c) A message will display if application is running on website correctly

```

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux
/zmscxwe28ls6omp4fnfdu1uyu
GiancarloCedano@Giancarlos-MacBook-Pro Prenova % docker run --rm -p 8005:8003 pr
enova-app
INFO:      Started server process [1]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
INFO:      Uvicorn running on http://0.0.0.0:8003 (Press CTRL+C to quit)
Starting Prenova...
Bound server host: 0.0.0.0:8003
Open in browser: http://127.0.0.1:8003

```

- 6) Once application is running, open up Google Chrome and load up website
 - a) Type in the following website to load the application
 - i) <http://localhost:8005/>
- 7) Once application is loaded, navigate tabs and change display settings as desired
 - a) Display setting change (dark mode) available on top right corner of application



For Testing

Testing Actual Patient Data (XG Boosted Model Scoring)

- 1) Load up website as stated in previous section
- 2) Load up Folder Containing Testing Demo CSV Files
 - a) Folder Labeled "Presentation Demos"
- 3) In the Upload Dropbox on the Vitals and Lab Input tab of the website, enter CSV file desired
 - a) Low Risk
 - i) Pos
 - ii) neg
 - b) High Risk
 - i) Pos
 - ii) Neg

Upload Structured Patient CSV

Upload CSV

Browse...

No file selected

- 4) View Updated Score on Alert Banner (top right) and within the Risk Assessment Tab

- **This automatically runs the CSV file inputs within the XG Boosted Tree Model for Scoring**



Name

John Doe

Age

54

Sex

Male

Hospital ID

A10298

Vitals & Labs Input

Risk Assessment

Top Contributors

Other Vitals / Labs

Lab Table

Sepsis Risk Score

HIGH RISK

Score: 0.84

Scoring: Demo override scoring (13/13 top predictors present)

Risk Level: HIGH (Clinical Fallback Score: 0.841 | 13/13 top predictors present)

HIGH RISK DETECTED

Clinical Fallback Score: 0.84

Scoring: Demo override scoring (13/13 top predictors present)

Please review patient vitals and consider clinical intervention.

Testing Actual Patient Data (Clinical Fallback Scoring)

- 1) Load up website as stated in previous section
- 2) Load up Folder Containing Testing Demo CSV Files
- 3) In the Upload Dropbox on the Vitals and Lab Input tab of the website, enter CSV file desired
 - a) Low Risk
 - i) Pos
 - ii) neg
 - b) High Risk
 - i) Pos

ii) Neg

Upload Structured Patient CSV

Upload CSV

Browse...

No file selected

- 4) On the “Vitals and Lab Input” Tab of the website, click on the slider to the right of the “Use Demo Override Scoring” Section to toggle between Clinical Fallback Score and XG Boosted Model

a) Model Used stated in Banner

Vitals & Labs Input Risk Assessment Top Contributors Other Vitals / Labs Lab Table

Upload Structured Patient CSV

Upload CSV

Browse... Highest Risk_pos.csv

Upload complete

Active row used across all tabs

1

Provider Overrides

Use demo override scoring
Prioritizes the clinical fallback score so override tests can visibly change LOW / MODERATE / HIGH risk.

HIGH RISK DETECTED
Clinical Fallback Score: 0.84
Scoring: Demo override scoring (13/13 top predictors present)
Please review patient vitals and consider clinical intervention.

Testing Extreme Cases (Very Low or Very High Risk)

- 1) Load up website as stated in previous section
- 2) Load up Folder Containing Testing Demo CSV Files
- 3) In the Upload Dropbox on the Vitals and Lab Input tab of the website, enter CSV file desired
 - a) Low Risk
 - i) Pos
 - ii) neg
 - b) High Risk
 - i) Pos
 - ii) Neg
- 4) On the “Vitals and Lab Input” Tab of the website, click on the slider to the right of the “Use Demo Override Scoring” Section to switch to XG Boosted Model
 - a) **Only Clinical Fallback Scoring allows for extreme cases**
- 5) Enter the following information for desired extreme in “Overrides” Test Input Box and click Apply Overrides After
 - a) Low Risk Drastic Change:
 - i) ast_bin3_mean: 20
 - ii) alt_bin3_mean: 20
 - iii) wbc_blood_bin3_mean: 6

- iv) pulse_bin3_mean: 70
 - v) temp_bin3_mean: 36.8
 - vi) lactate_bin3_mean: 1.0
 - vii) bili_tot_bin3_mean: 0.5
 - viii) bili_dir_bin3_mean: 0.1
 - ix) admission_type_elective: 1
 - x) admission_type_emergency: 0
 - xi) admission_type_newborn: 0
 - xii) admission_type_urgent: 0
 - xiii) emb_81: 0.05
 - xiv) emb_545: 0.05
- b) High Risk Drastic Change:
- i) ast_bin3_mean: 250
 - ii) alt_bin3_mean: 220
 - iii) wbc_blood_bin3_mean: 22
 - iv) pulse_bin3_mean: 135
 - v) temp_bin3_mean: 39.5
 - vi) lactate_bin3_mean: 5.0
 - vii) bili_tot_bin3_mean: 6.0
 - viii) bili_dir_bin3_mean: 2.0
 - ix) admission_type_elective: 0
 - x) admission_type_emergency: 1
 - xi) admission_type_newborn: 0
 - xii) admission_type_urgent: 0
 - xiii) emb_81: 1.0
 - xiv) emb_545: 1.0

Vitals & Labs Input

Risk Assessment

Top Contributors

Other Vitals / Labs

Lab Table

Upload Structured Patient CSV

Upload CSV

Browse...

Highest Risk_pos.csv

Upload complete

Active row used across all tabs

1

Provider Overrides

Use demo override scoring

Prioritizes the clinical fallback score so override tests can visibly change LOW / MODERATE / HIGH risk.

HIGH RISK DETECTED

Clinical Fallback Score: 0.84

Scoring: Demo override scoring (13/13 top predictors present)

Please review patient vitals and consider clinical intervention.